

# IAFMNet: Information-Aware Feature Modulation for Efficient Super-Resolution

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## Abstract

Single Image Super-Resolution (SISR) aims to reconstruct a high-resolution (HR) image from a low-resolution (LR) input, a task that becomes increasingly challenging under real-world computational constraints. However, most efficient SISR methods adopt lightweight, spatially uniform strategies that allocate equal computation and attention across all regions, ignoring the uneven distribution of visual complexity. From an information-theoretic perspective, textures and edges inherently carry more critical information, resulting in reconstruction errors that are disproportionately concentrated in these regions. This motivates the allocation of greater computational resources and attention to these informative areas. In this paper, we propose IAFMNet, an Information-Aware Feature Modulation network for efficient SR. At its core lies the Information Density Map (IDM), which is estimated in an unsupervised manner by minimizing the Information Entropy Loss, thereby highlighting informative regions with high estimated encoding costs. Guided by the IDM, IAFMNet adopts a synergistic dual-branch design: (1) a sparse convolution branch that dynamically allocates computation to informative areas while bypassing low-information regions, and (2) an implicit modulation branch that adaptively emphasizes complex regions through information-aware affine transformations. Extensive experiments demonstrate that IAFMNet effectively identifies informative regions and achieves superior visual fidelity with reduced computational overhead.

## 1. Introduction

Single Image Super-Resolution (SISR), a long-standing and challenging task in low-level computer vision, aims to reconstruct high-resolution (HR) images from their low-resolution (LR) counterparts [18, 37]. Fueled by the proliferation of ultra-high-definition displays and streaming media applications, this ill-posed task has garnered significant

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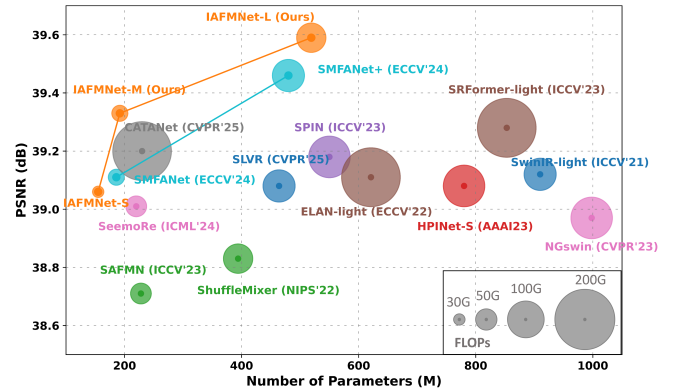


Figure 1. Performance and model complexity comparison on the Manga109 [22] dataset for  $\times 4$  super-resolution.

research interest. However, the inherent computational and memory constraints of real-world platforms necessitate the development of efficient SR networks that can deliver superior visual quality without incurring prohibitive costs.

In the pursuit of efficiency, recent research has typically focused on globally efficient network architectures. Within the CNN paradigm, this has led to a variety of strategies, such as developing more efficient convolution operators [8, 23, 28], mining architectural experts [32], or designing sophisticated feature modulation schemes [29, 38]. A similar trend is also observed in Transformer-based methods. While ViTs excel at capturing long-range dependencies, their inevitable quadratic complexity has necessitated various lightweight designs. These solutions range from partitioning features into predefined, content-agnostic windows [17, 36, 39] to more advanced, content-aware strategies [20, 34, 35] that aggregate semantically similar tokens. Though architecturally diverse, these approaches share a common philosophy: an efficient but static strategy is applied uniformly across all pixels and regions.

However, the validity of this spatially uniform paradigm becomes questionable when analyzing the SR task itself. Downsampling disproportionately degrades an image's high-frequency details and edges, the very regions



Figure 2. Illustration of the difference map (d) between the HR image and the SR image (b) reconstructed by baseline  $f(\cdot)$  on Urban100. The estimated IDM  $\theta$  in (e), obtained via  $\mathcal{L}_{IE}$ , successfully captures the informative regions highlighted in the difference map, and (c) shows the superior reconstruction result produced by our proposed IAFMNet under the guidance of  $\theta$ .

that, paradoxically, carry the most critical information [24]. The weakened representation of these areas in the LR input, in turn, leads to unfaithful reconstructions and unpleasant artifacts in SISR. As depicted in Fig. 2(d), the absolute difference map between a ground-truth (GT) image and an SR result reveals a distinct spatially varying error distribution, where regions with complex textures consistently exhibit significantly higher reconstruction errors. This observation motivates a critical question: *Should these reconstruction-challenging regions be allocated the same computational resources as easily reconstructed areas, especially under stringent computational constraints?* We thus argue for a new paradigm that prioritizes computation in these visually complex regions, which are often associated with high information density.

To address this, we propose an Information-Aware Feature Modulation Network (IAFMNet), which introduces a principled approach to dynamic computational allocation for efficient SISR. Inspired by information theory [25], we first estimate a pixel-wise Information Density Map (IDM) in an unsupervised manner by minimizing the Information Entropy Loss, i.e., the encoding cost of quantized features under the learned entropy model. As validated in Fig. 2(e), the resulting IDM provides interpretable guidance and successfully highlights reconstruction-challenging regions corresponding to high information density. The IDM then directs a dual-branch, information-guided feature enhancement strategy that consists of an Information-Guided Resource Allocation (IGRA) branch that prioritizes computation in informative regions identified by the IDM, and an Affine Recalibration Module (ARM) that performs fine-grained feature recalibration under the guidance of the IDM. This combination of hard computational allocation and soft feature refinement enables our method to achieve a superior trade-off between reconstruction performance and computational efficiency, as shown in Fig. 1. Our main contributions are summarized as follows:

- We propose to estimate a pixel-wise Information Density Map (IDM) by minimizing an unsupervised Information Entropy Loss ( $\mathcal{L}_{IE}$ ), which provides a principled guid-

ance for identifying informative regions for SISR. To the best of our knowledge, this is the first work to enhance degraded features for SISR from the perspective of information density.

- Under the guidance of the learned IDM, we introduce a novel Information-Aware hybrid Feature Modulation Network (IAFMNet) that synergistically combines explicit, hard resource allocation (via the IGRA branch with sparse convolution) and implicit, soft feature modulation (via the ARM branch).
- Extensive experiments demonstrate that the proposed IAFMNet achieves a superior trade-off between model complexity and reconstruction performance on public benchmarks, validating the effectiveness of our information-guided strategy.

## 2. Related Work

### 2.1. Deep Network-Based Super-Resolution

**CNN-based Methods.** Pioneered by SRCNN [8], Convolutional Neural Networks (CNNs) have long been the dominant paradigm in SISR. Key innovations like residual learning [10], introduced in VDSR [14] and further refined in EDSR [18], enabled the training of much deeper and larger networks, leading to significant performance gains. Further advancements, such as the residual-in-residual structure and channel attention in RCAN [37], pushed performance to new limits, although RCAN has 15.59M parameters and over 400 layers. However, the high computational and memory demands of these large-scale models make them impractical for many real-world applications.

**Transformer-based Methods.** More recently, Vision Transformers (ViTs) have emerged as a powerful alternative, leveraging self-attention to model the long-range dependencies that CNNs traditionally struggle to capture. While models such as IPT [6] first demonstrated the effectiveness of Transformers for SR, their utility was constrained by the quadratic complexity of global self-attention. Influential baselines like SwinIR [17] addressed this issue by confining computation to local,

non-overlapping windows. Other methods, including ELAN [36] and ESRT [21], explore group-wise attention or feature downscaling to reduce complexity. Despite their promising performance, these ViT-based SR methods often remain computationally intensive.

## 2.2. Efficient SR Methods

Early efficient super-resolution models mainly improved efficiency through architectural design, like post-upsampling [8, 27] and information distillation [13]. More recent CNN-based approaches have further advanced efficiency through lightweight feature modulation [29, 38], expert-mining schemes [32], and efficient blueprint convolution [23]. Lightweight Vision Transformers (ViTs) pursue a similar goal by constraining self-attention within predefined windows [17, 36, 39] or grouping semantically similar tokens [20, 34, 35]. Despite these improvements, most methods still adopt globally uniform computational strategies that allocate equal computational resources to all regions, ignoring the highly uneven distribution of reconstruction complexity. To address this imbalance, several studies have explored complexity-aware resource allocation based on data characteristics, such as empirical PSNR gaps [15], localized gradients [11], or adaptively learned spatial attention [31]. These methods demonstrate the benefit of non-uniform computation, but their complexity estimates typically rely on empirical heuristics or coarse proxy signals, limiting their ability to characterize reconstruction difficulty in a principled manner. A critical gap therefore remains in principled and adaptive computation allocation.

## 3. Methodology

The overall architecture of the proposed IAFMNet is illustrated in Fig. 3(a). It consists of four key components: a shallow feature extractor, an Information Density Estimator (IDE), a series of Information-guided Feature Enhancement Blocks (IFEBS), and an image reconstruction module. Given an input LR image  $y$ , a single  $3 \times 3$  convolutional layer first extracts shallow features  $F_0$ . These features are then processed by the Information Density Estimator (IDE), which produces two critical outputs in parallel: a mean feature map  $\mu$ , later used as residual features, and the core guidance signal of our framework, the Information Density Map (IDM)  $\theta$ . The original shallow features  $F_0$ , along with this shared IDM  $\theta$ , are then propagated through a stack of  $N$  IFEBS. Each IFEBS contains two sub-modules: an IFE Module (IFEM) and a Channel-Gated Feed-forward Network (CGFN) for progressive feature refinement. After the final IFEBS, the refined features are fused with the mean map  $\mu$ . Finally, a lightweight convolutional layer followed by a PixelShuffle operation reconstructs the final HR image  $\hat{x}$ .

### 3.1. Information Density Estimator

The spatial downsampling inherent in SISR can be conceptualized as a low-pass filter that disproportionately attenuates high-frequency information, which often corresponds to informative textures and edges. Faithfully reconstructing these heavily degraded regions is therefore essential for high-quality SR. To this end, we identify and prioritize these regions from a novel information-theoretic perspective. Given a signal  $x$  with probability  $p(x)$ , the amount of information it carries can be expressed as:

$$I(x) = -\log_2 p(x). \quad (1)$$

This principle [26] suggests that complex and unpredictable signals, which occur with lower probability, inherently contain more information. According to Shannon entropy theory [25], the theoretical minimum number of bits required to encode a signal (its rate cost  $R$ ) is determined by its probability distribution. Specifically, for a discretized signal  $\hat{x}$  drawn from a true but unknown data distribution  $m(\hat{x})$ , the expected rate cost  $R$  is formally defined as the cross-entropy between  $m$  and a learned, approximate distribution  $p_{\hat{x}}(\hat{x})$ :

$$R = \mathbb{E}_{\hat{x} \sim m}[-\log_2 p_{\hat{x}}(\hat{x})] = H(m) + D_{KL}(m||p_{\hat{x}}), \quad (2)$$

where  $H(m)$  denotes the entropy of the true distribution and  $D_{KL}(\cdot)$  denotes the KL divergence. Minimizing this cost encourages the learned distribution  $p_{\hat{x}}$  to better approximate the true distribution  $m$ . In optimal compression, resources are allocated adaptively according to the information density of the signal: complex regions require more bits for faithful representation, whereas redundant regions can be encoded more compactly to improve efficiency. Motivated by [5], we further interpret the rate cost as a theoretically grounded indicator of reconstruction difficulty in SISR.

Within the shallow features  $F_0 \in \mathbb{R}^{h \times w \times C}$  extracted from the LR image, the occurrence probability of these heavily attenuated high-frequency textures and details is significantly lower than that from smooth regions. This motivates us to employ the information density as a principled measure to characterize and prioritize these informative regions for enhancement. To this end, we first model the probability distribution of a feature map  $F$ . To bridge continuous features and discrete encoding cost [3], we approximate differentiable quantization by injecting uniform noise during training:  $\hat{F} = Q(F) = F + \mathcal{U}(-\frac{1}{2}, \frac{1}{2})$ .

Subsequently, we model the distribution of the quantized feature map  $\hat{F}$  with a fully factorized density model, where each feature element  $\hat{F}_i$  is assumed to follow an independent Gaussian distribution parameterized by mean  $\mu_i$  and scale  $\theta_i$ . Under this assumption, the joint probability distribution for  $\hat{F}$  can be written as  $p_{\hat{F}}(\hat{F} | \mu, \theta) = \prod_i p_{\hat{F}_i}(\hat{F}_i | \mu_i, \theta_i)$ . With this distribution model established, the likelihood of observing each quantized element  $\hat{F}_i$  is given by

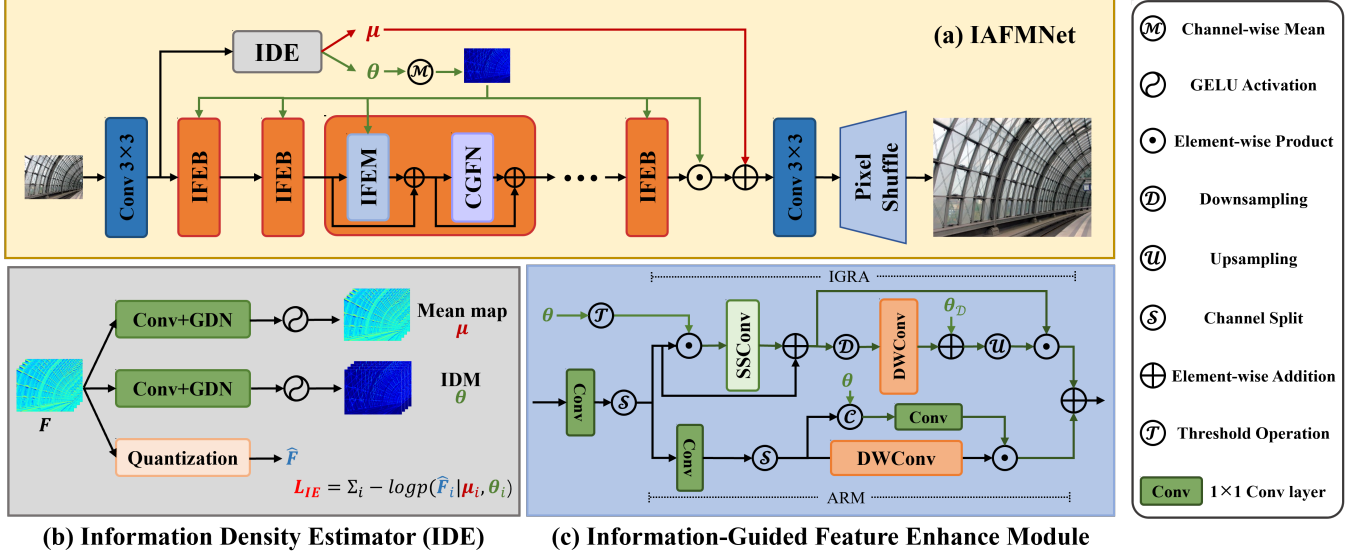


Figure 3. Illustration of the proposed IAFMNet: (a) the overall network architecture, (b) the information density estimator, and (c) the information-guided feature enhancement module.

the integral of the continuous probability density function over the quantization bin,  $[\hat{F}_i - \frac{1}{2}, \hat{F}_i + \frac{1}{2}]$ . This integral can be efficiently computed as:

$$\begin{aligned}
 p_{\hat{F}_i}(\hat{F}_i | \mu_i, \theta_i) &= \int_{\hat{F}_i - \frac{1}{2}}^{\hat{F}_i + \frac{1}{2}} \mathcal{N}(x | \mu_i, \theta_i^2) dx \\
 &= \Phi\left(\frac{\hat{F}_i + \frac{1}{2} - \mu_i}{\theta_i}\right) - \Phi\left(\frac{\hat{F}_i - \frac{1}{2} - \mu_i}{\theta_i}\right),
 \end{aligned} \quad (3)$$

where  $\Phi(\cdot)$  denotes the cumulative distribution function of the standard Gaussian distribution. Here,  $\mu$  and  $\theta$  are predicted from the original feature map  $F$  by two dedicated CNN-based modules, each consisting of a convolution layer followed by Generalized Divisive Normalization (GDN) [3], which is designed for density modeling.

Finally, the element-wise encoding cost  $R_{\hat{F}_i}$  can be obtained via its negative log-likelihood, and the total Information Entropy Loss ( $\mathcal{L}_{IE}$ ) is the sum of the per-pixel encoding costs:

$$\mathcal{L}_{IE} = \sum_i R_{\hat{F}_i} = \sum_i -\log_2 p_{\hat{F}_i}(\hat{F}_i | \mu_i, \theta_i). \quad (4)$$

By minimizing  $\mathcal{L}_{IE}$ , the network is encouraged to learn an accurate density model and produce a spatially varying scale map  $\theta$ . This learned map serves as our Information Density Map (IDM), providing a principled and interpretable signal to characterize reconstruction difficulty and guide subsequent feature enhancement.

### 3.2. Information-guided Feature Enhancement Module

To prioritize the informative regions identified by the IDM, we design the Information-guided Feature Enhancement Module (IFEM). As illustrated in Fig. 3(c), IFEM introduces a hybrid computational strategy, processing features through two parallel, synergistic branches: 1. the Information-Guided Resource Allocation (IGRA) branch for explicit, hard computational allocation, and 2. the Affine Recalibration Module (ARM) for implicit, soft feature modulation. Given an input feature map  $F_{in}$ , it is first passed through a convolution layer for channel expansion. The obtained feature is then split into two parts,  $F_1$  and  $F_2$ , which are fed into the IGRA and ARM branches, respectively:

$$F_1, F_2 = \mathcal{S}(\text{Conv}(F_{in})), \quad F'_2 = \text{ARM}(F_2), \quad F'_1 = \text{IGRA}(F_1), \quad (5)$$

where  $\mathcal{S}(\cdot)$  denotes a channel-wise split operation. The outputs of the two branches,  $F'_1$  and  $F'_2$ , are then fused by element-wise addition to produce the enhanced feature.

### 3.3. Information-Guided Resource Allocation

As discussed in the previous section, natural images exhibit spatially uneven information distribution, where textures and edges contain significantly richer information than flat areas. Conventional CNNs, however, adopt a global and uniform convolution strategy, resulting in redundant computation in information-poor regions. To address this, we propose the IGRA, which adaptively allocates computation to informative regions identified by the learned IDM  $\theta$ .

Specifically, we first generate a binary mask  $\mathbf{M} \in \{0, 1\}^{H \times W}$  by identifying the top- $k\%$  most informative

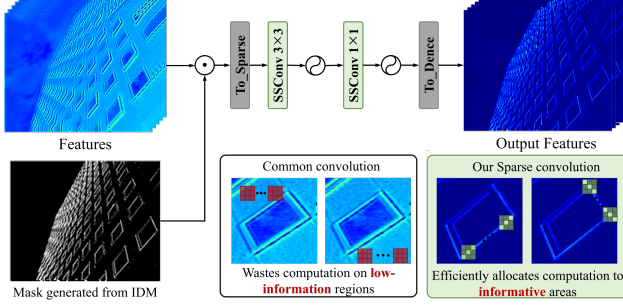


Figure 4. Illustration of the proposed information-aware sparse convolution module.

spatial locations from IDM  $\theta$ :  $\mathbf{M} = \mathcal{T}(\theta, k)$ , where  $k$  is a predefined sparsity threshold. This mask creates a sparse feature map:  $\mathbf{F}_{\text{sparse}} = \mathbf{M} \odot \mathbf{F}_1$ .

This sparse representation is then processed by a series of Submanifold Sparse Convolution (SSC) layers [9] to adaptively allocate computing resources. Unlike standard sparse convolution, which dilates the active region, SSC is specifically designed to restrict computation only to informative sites defined by  $\mathbf{M}$ , preserving the exact sparsity pattern throughout the operation. The output feature  $\mathbf{F}_{\text{ssc}}$  at position  $p$  can be calculated by:

$$\mathbf{F}_{\text{ssc}}(p) = \begin{cases} \sum_{q \in \mathcal{N}(p)} \mathbf{W}(q-p) \cdot \mathbf{F}_{\text{sparse}}(q) + \mathbf{F}_1(p), & \text{if } \mathbf{M}(p) = 1, \\ \mathbf{F}_1(p), & \text{otherwise.} \end{cases} \quad (6)$$

where  $\mathcal{N}(p)$  and  $\mathbf{W}(q-p)$  denote the neighborhood of the location  $p$  and the weight of the convolution kernel.

As shown in Fig. 4, this design ensures that computation is concentrated on informative regions, while smooth or redundant areas are skipped, significantly reducing FLOPs. To compensate for the potential drawbacks of the hard-thresholding operation, we incorporate a lightweight self-attention mechanism to refine the sparse output. Specifically, the attention map  $\mathbf{A}$  is also guided by the IDM, generated by fusing the downsampled feature map ( $\mathbf{F}_{\mathcal{D}}$ ) of  $\mathbf{F}_{\text{ssc}}$  and the downsampled IDM ( $\theta_{\mathcal{D}}$ ):

$$\mathbf{A} = \text{Upsample}(\text{Conv}_{1 \times 1}(\mathbf{F}_{\mathcal{D}} + \theta_{\mathcal{D}})) \quad (7)$$

The resulting attention map  $\mathbf{A}$  is then used to refine the sparse convolution output:  $\mathbf{F}_{\text{refined}} = \mathbf{A} \odot \mathbf{F}_{\text{ssc}}$ .

### 3.4. Affine Recalibration Module

As a complement to the explicit resource allocation in IGRA, the Affine Recalibration Module implicitly embeds the guidance of the IDM through a soft, channel-wise feature modulation mechanism.

Given the input feature map  $\mathbf{F}_2 \in \mathbb{R}^{H \times W \times C}$ , a  $1 \times 1$  convolution is first applied to expand the channel dimension, after which the expanded feature is split into two equal

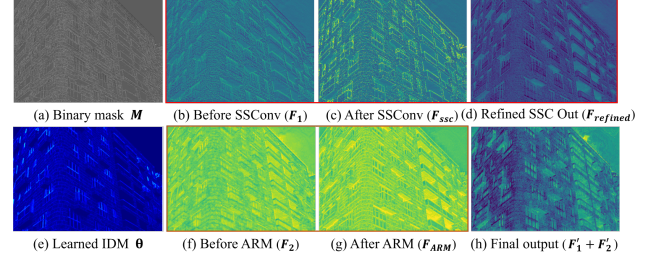


Figure 5. Visualization of progressive feature enhancement via IDM-guided sparse allocation and soft modulation.

parts. The first part is concatenated with the IDM  $\theta$  along the channel dimension and passed through a  $1 \times 1$  convolution to generate the modulation parameters  $s$ :

$$s = \text{Conv}_{1 \times 1}([\mathcal{S}(\text{Conv}_{1 \times 1}(\mathbf{F}_2))[0], \theta]). \quad (8)$$

Meanwhile, the second part is processed using a lightweight depth-wise convolution to efficiently capture local structural features:

$$\mathbf{F}_{\text{local}} = \text{DWConv}(\mathcal{S}(\text{Conv}_{1 \times 1}(\mathbf{F}_2))[1]). \quad (9)$$

Finally, the local features are recalibrated by the information-aware scales, producing the branch output:  $\mathbf{F}_{\text{ARM}} = \mathbf{F}_{\text{local}} \odot s$ .

## 4. Experiments

### 4.1. Experimental Setting

**Dataset.** To ensure a fair and comprehensive comparison, we follow the standard training and evaluation protocols in [38]. We train all models on the widely used DF2K dataset, which combines DIV2K and Flickr2K [30]. The LR training images are generated by bicubic downsampling. For performance evaluation, we compare our method with other competing methods on five well-established benchmark datasets, namely Set5 [4], Set14 [33], BSD100 [2], Urban100 [12], and Manga109 [22].

**Evaluation metrics.** We adopt Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) as our quantitative evaluation metrics. Following standard practice [18], we calculate PSNR and SSIM on the luminance channel (Y-channel) of images converted to the YCbCr color space to assess reconstruction quality.

**Implementation details.** During training, we perform data augmentation by extracting  $64 \times 64$  LR patches and applying random horizontal flips and 90-degree rotations. Our model is trained under the supervision of a composite loss function  $\mathcal{L} = \mathcal{L}_1 + \lambda \mathcal{L}_{IE}$ , which combines the standard  $\mathcal{L}_1$  pixel loss with the proposed Information Entropy Loss ( $\mathcal{L}_{IE}$ ) as defined in Eq. 4. The network is optimized using the Adam optimizer with a batch size of 64. The initial learning rate is set to  $1 \times 10^{-3}$  and annealed to  $1 \times 10^{-5}$

| Scale      | Methods            | Params | FLOPs | Set5                | Set14               | BSD100              | Urban100            | Manga109            |
|------------|--------------------|--------|-------|---------------------|---------------------|---------------------|---------------------|---------------------|
| $\times 2$ | FSRCNN [8]         | 12K    | 6G    | 37.00/0.9558        | 32.63/0.9088        | 31.53/0.8920        | 29.88/0.9020        | 36.67/0.9694        |
|            | EDSR-baseline [18] | 1370K  | 316G  | 37.99/0.9604        | 33.57/0.9175        | 32.16/0.8994        | 31.98/0.9272        | 38.54/0.9769        |
|            | CARN-M [1]         | 415K   | 91G   | 37.53/0.9583        | 33.26/0.9141        | 31.92/0.8960        | 31.23/0.9193        | -/-                 |
|            | IMDN [13]          | 694K   | 159G  | 38.00/0.9605        | 33.63/0.9177        | 32.19/0.8996        | 32.17/0.9283        | 38.88/0.9773        |
|            | ShuffleMixer [28]  | 394K   | 91G   | 38.01/0.9606        | 33.63/0.9180        | 32.17/0.8995        | 31.89/0.9257        | 38.83/0.9774        |
|            | LAPAR-A [16]       | 548K   | 137G  | 38.01/0.9605        | 33.62/0.9183        | 32.19/0.8999        | 32.10/0.9283        | 38.67/0.9772        |
|            | SMFANet [38]       | 186K   | 41G   | 38.08/0.9607        | 33.65/0.9185        | 32.22/0.9002        | 32.20/0.9282        | 39.11/0.9779        |
|            | SeemoRe-T [32]     | 220K   | 45G   | 38.06/0.9608        | 33.65/0.9186        | 32.23/0.9004        | 32.22/0.9286        | 39.01/0.9771        |
|            | SAFMN [29]         | 228K   | 52G   | 38.00/0.9605        | 33.54/0.9177        | 32.16/0.8995        | 31.84/0.9256        | 38.71/0.9771        |
|            | IAFMNet(ours)      | 198K   | 42G   | <b>38.11/0.9611</b> | <b>33.72/0.9190</b> | <b>32.28/0.9006</b> | <b>32.52/0.9312</b> | <b>39.32/0.9792</b> |
| $\times 3$ | FSRCNN [8]         | 12K    | 5G    | 33.16/0.9140        | 29.43/0.8242        | 28.53/0.7910        | 26.43/0.8080        | 30.98/0.9212        |
|            | EDSR-baseline [18] | 1555K  | 160G  | 34.37/0.9270        | 30.28/0.8417        | 29.09/0.8052        | 28.15/0.8527        | 33.45/0.9439        |
|            | CARN-M [1]         | 415K   | 46G   | 33.99/0.9236        | 30.08/0.8367        | 28.91/0.8000        | 27.55/0.8385        | -/-                 |
|            | IMDN [13]          | 703K   | 72G   | 34.36/0.9270        | 30.32/0.8417        | 29.09/0.8046        | 28.17/0.8519        | 33.61/0.9445        |
|            | ShuffleMixer [28]  | 415K   | 43G   | 34.40/0.9272        | 30.37/0.8423        | 29.12/0.8051        | 28.08/0.8498        | 33.69/0.9448        |
|            | LAPAR-A [16]       | 594K   | 114G  | 34.36/0.9267        | 30.34/0.8412        | 29.11/0.8054        | 28.15/0.8523        | 33.51/0.9441        |
|            | SMFANet [38]       | 191K   | 19G   | 34.42/0.9274        | 30.41/0.8430        | 29.16/0.8065        | 28.22/0.8523        | 33.96/0.9460        |
|            | SeemoRe-T [32]     | 225K   | 20G   | 34.46/0.9276        | 30.44/0.8445        | 29.15/0.8063        | 28.27/0.8538        | 33.92/0.9460        |
|            | SAFMN [29]         | 233K   | 23G   | 34.34/0.9267        | 30.33/0.8418        | 29.08/0.8048        | 27.95/0.8474        | 33.52/0.9437        |
|            | IAFMNet(ours)      | 208K   | 19G   | <b>34.53/0.9281</b> | <b>30.61/0.8454</b> | <b>29.20/0.8068</b> | <b>28.48/0.8561</b> | <b>34.17/0.9476</b> |
| $\times 4$ | FSRCNN [8]         | 12K    | 5G    | 30.71/0.8657        | 27.59/0.7535        | 26.98/0.7150        | 24.62/0.7280        | 27.90/0.8517        |
|            | EDSR-baseline [18] | 1518K  | 114G  | 32.09/0.8938        | 28.58/0.7813        | 27.57/0.7357        | 26.04/0.7849        | 30.35/0.9067        |
|            | CARN-M [1]         | 415K   | 33G   | 31.92/0.8903        | 28.42/0.7762        | 27.44/0.7304        | 25.62/0.7694        | -/-                 |
|            | ShuffleMixer [28]  | 411K   | 28G   | 32.21/0.8953        | 28.66/0.7827        | 27.61/0.7366        | 26.08/0.7835        | 30.65/0.9093        |
|            | LAPAR-A [16]       | 659K   | 94G   | 32.15/0.8944        | 28.61/0.7818        | 27.61/0.7366        | 26.14/0.7871        | 30.42/0.9074        |
|            | SMFANet [38]       | 197K   | 11G   | 32.25/0.8956        | 28.71/0.7833        | 27.64/0.7377        | 26.18/0.7862        | 30.82/0.9104        |
|            | SeemoRe-T [32]     | 232K   | 12G   | 32.31/0.8965        | 28.72/0.7840        | 27.65/0.7384        | 26.23/0.7883        | 30.82/0.9107        |
|            | SAFMN [29]         | 240K   | 14G   | 32.18/0.8948        | 28.60/0.7813        | 27.58/0.7359        | 25.97/0.7809        | 30.43/0.9063        |
|            | SLVR-S [23]        | 158K   | 8.5G  | 32.20/0.8952        | 28.65/0.7822        | 27.57/0.7361        | 26.08/0.7854        | 30.53/0.9084        |
|            | IAFMNet-S(ours)    | 166K   | 8.2G  | 32.24/0.8956        | 28.68/0.7828        | 27.61/0.7369        | 26.15/0.7857        | 30.71/0.9098        |
|            | IAFMNet(ours)      | 220K   | 11G   | <b>32.37/0.8974</b> | <b>28.80/0.7844</b> | <b>27.68/0.7381</b> | <b>26.39/0.7891</b> | <b>30.97/0.9121</b> |

Table 1. Comparison with SOTA lightweight SR methods on public benchmark datasets. Best results are colored with red.

| Methods             | Params | FLOPs | Set5                | Set14               | Urban100            | Manga109            |
|---------------------|--------|-------|---------------------|---------------------|---------------------|---------------------|
| ESRT [21]           | 752K   | 298G  | 32.19/0.8947        | 28.69/0.7833        | 26.39/0.7962        | 30.75/0.9100        |
| SwinIR-light [17]   | 930K   | 65G   | 32.44/0.8976        | 28.77/0.7858        | 26.47/0.7980        | 30.92/0.9151        |
| ELAN-light [36]     | 640K   | 54G   | 32.43/0.8975        | 28.78/0.7858        | 26.54/0.7982        | 30.92/0.9150        |
| HPINet-S [19]       | 463K   | 88G   | 32.47/0.8987        | 28.80/0.7872        | 26.59/0.8016        | 30.92/0.9143        |
| NGswin [7]          | 1019K  | 40G   | 32.33/0.8963        | 28.78/0.7859        | 26.45/0.7963        | 30.80/0.9128        |
| SPIN [34]           | 555K   | 42G   | 32.48/0.8983        | 28.80/0.7862        | 26.55/0.7998        | 30.98/0.9156        |
| SRFormer-light [39] | 873K   | 63G   | 32.51/0.8988        | 28.82/0.7872        | 26.67/0.8032        | 31.17/0.9165        |
| SMFANet+ [38]       | 496K   | 28G   | 32.51/0.8985        | 28.87/0.7872        | 26.56/0.7976        | 31.29/0.9163        |
| CATANet [20]        | 535K   | 41G   | <b>32.58/0.8998</b> | <b>28.90/0.7880</b> | <b>26.87/0.8081</b> | <b>31.31/0.9183</b> |
| IAFMNet-L (ours)    | 519K   | 28G   | <b>32.57/0.8993</b> | <b>28.92/0.7882</b> | <b>26.73/0.8038</b> | <b>31.39/0.9173</b> |

Table 2. Performance comparison against lightweight CNN- and ViT-based SR methods on public benchmark datasets. Best and second-best results are colored with red and blue.

via a cosine annealing schedule over 1,000,000 iterations. All experiments are conducted in the PyTorch framework on two NVIDIA GeForce RTX 4090 GPUs.

## 4.2. Comparison with State-of-the-Art Methods

We develop three variants of our model: IAFMNet-S (166K Params), IAFMNet (220K Params), and IAFMNet-L (519K Params), and compare them with a range of lightweight CNN-based SR approaches, including FSRCNN [8], EDSR-baseline [18], CARN-M [1], IMDN [13], ShuffleMixer [28], LAPAR-A [16], SMFANet [38], SeemoRe-T [32], SAFMN [29], and SLVR-S [23]. We further compare the large version IAFMNet-L with SMFANet+ [38]

and the recently proposed lightweight ViT-based methods, including CATANet [20], SRformer-light [39], ELAN-light [36], and SPIN [34]. The quantitative comparison results for upscaling factors of  $\times 2$ ,  $\times 3$ , and  $\times 4$  on five benchmark datasets are reported in Table 1, together with the number of parameters (Params) and FLOPs to evaluate model efficiency. To ensure a fair comparison across different computational budgets, all FLOPs are uniformly calculated for a  $1280 \times 720$  HR output.

**Quantitative comparison.** As shown in Table 1, our proposed IAFMNet outperforms competing methods across nearly all datasets and scaling factors, while maintaining a lower or comparable number of parameters and FLOPs. The advantage is particularly pronounced on complex datasets. For instance, on the Urban100 for  $\times 2$  SR, our method achieves a significant 0.30 dB PSNR improvement over SeemoRe-T while utilizing fewer FLOPs. Furthermore, even compared against recent state-of-the-art models such as SLVR-S, our method consistently exhibits superior performance. Table 2 presents a comparison under a larger computational budget. Our larger model, IAFMNet-L, achieves competitive or even superior performance against

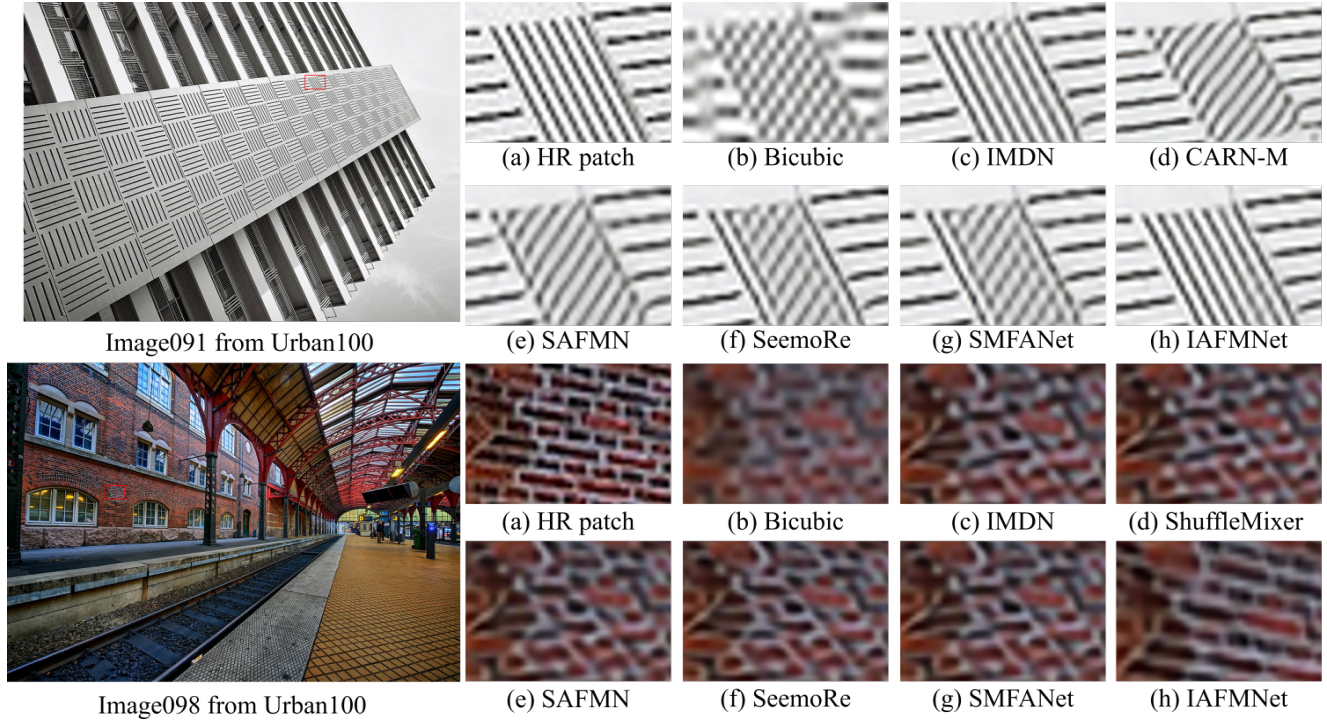


Figure 6. Visual comparison for SISR on the Urban100 dataset.

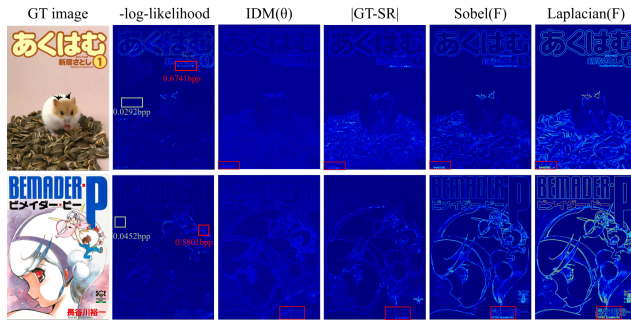


Figure 7. Visual comparison of the proposed Information Density Map (IDM) with traditional gradient-based operators (Sobel and Laplacian). The IDM effectively highlights information-rich textures that are often overlooked by simple edge detectors.

ViT-based methods despite its lower model complexity. For example, compared with SRFormer-light [39], IAFMNet-L uses substantially fewer parameters and FLOPs (519K vs. 873K parameters and 28G vs. 63G FLOPs), while achieving higher PSNR on all reported datasets. These improvements can be attributed to our principled, information-guided strategy. By employing the IDM to identify and prioritize inherently informative regions that are difficult to reconstruct, our network allocates computational resources more effectively than the spatially uniform operations commonly used in both CNNs and ViTs.

**Qualitative Results.** We compare the visual results of our proposed IAFMNet with competing efficient SISR

methods on Urban100 for  $\times 4$  SR. As shown in Fig. 6, earlier lightweight methods such as IMDN and ShuffleMixer suffer from blurring and fail to restore sharp edges and parallel lines. While recent SOTA models (e.g., SAFMN, SeemoRe-T, SMFANet) perform better, they still exhibit noticeable artifacts, such as distorted structures in ‘img\_049’ or the failure to reconstruct fine grid patterns in ‘img\_098’. In contrast, our IAFMNet successfully restores more accurate parallel patterns and sharper boundaries, producing results that are visually more faithful to the ground truth. This visual superiority, especially in texture-rich regions, further supports the effectiveness of the IDM in guiding the network to focus on these informative regions.

### 4.3. Superiority over Traditional Gradient Operators

To demonstrate the effectiveness of our Information Density Map (IDM), we visualize the IDM compared with traditional gradient-based edge detectors, namely Sobel and Laplacian, in Fig. 7. As illustrated, traditional operators predominantly emphasize strong local boundaries, responding intensely to sharp contours and text outlines. Although visually salient, these simple edges are relatively easy to reconstruct in SISR. Furthermore, gradient-based methods struggle to capture dense, low-contrast details, such as fur or cluttered backgrounds, which suffer severe information loss. In contrast, our IDM is derived through entropy-based modeling, enabling it to go beyond simple gradient mag-

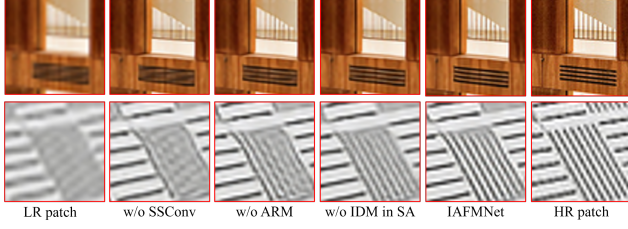


Figure 8. Visual comparison of the ablation study results. Zoom in for better visualization.

| $\lambda$ | Network | Params | FLOPs | Urban100     | Manga109     |
|-----------|---------|--------|-------|--------------|--------------|
| 0.001     | C+G     | 198K   | 42G   | 32.42/0.9291 | 39.18/0.9783 |
| 0.0001    | C+G     | 198K   | 42G   | 32.52/0.9312 | 39.32/0.9792 |
| 0.00001   | C+G     | 198K   | 42G   | 32.48/0.9303 | 39.26/0.9786 |
| 0.0001    | C       | 198K   | 41G   | 32.35/0.9287 | 39.11/0.9779 |
| 0.0001    | C+G+C+G | 211K   | 48G   | 32.55/0.9317 | 39.37/0.9802 |
| -         | Sobel   | 189K   | 39G   | 32.37/0.9290 | 39.14/0.9780 |

Table 3. Ablation studies on different IDE settings, where C and G denote a convolutional layer and a GDN layer, respectively.

nitide and better reflect the local encoding complexity of quantized features. It actively suppresses predictable edges and places significantly greater emphasis on intricate, unpredictable textures. By aligning more closely with actual reconstruction difficulty, the IDM provides a more reliable and principled guide for adaptive feature enhancement and computational resource allocation.

#### 4.4. Ablation Studies

We conduct extensive ablation experiments to verify the effectiveness and design choices of each proposed component in IAFMNet. Quantitative results are reported in Tables 3-5, and visual comparisons are provided in Fig. 8. All models are trained and evaluated on the  $\times 2$  SR task.

**Effectiveness of the IDE.** To evaluate the contribution of our Information Density Estimator, we analyze its key settings in Table 3. Since the full loss is  $\mathcal{L}_1 + \lambda \mathcal{L}_{IE}$ , we first study the impact of  $\lambda$ . The results show that setting  $\lambda$  to 0.0001 yields the best performance, bringing a notable gain of 0.06–0.14 dB on Manga109 compared with the other settings. Furthermore, we find that a simple IDE composed of one convolution layer and one GDN layer (C+G) provides the best efficiency. While a deeper C+G+C+G network offers a marginal performance gain (+0.05 dB on Manga109), it comes at a higher computational cost. Conversely, removing the GDN layer (C only), which is essential for density modeling [3], causes a significant performance drop of 0.17 dB. We also compared our IDM with the Sobel edge detector. Unlike a simple edge detector, our information-oriented IDM is learned through entropy-based modeling to highlight regions with high local encoding complexity, rather than merely strong gradients. This principled approach proves effective, as the IDM-guided model surpasses the Sobel-guided model by 0.15 dB on Urban100 and 0.18 dB on Manga109, demonstrating the superiority of our learned,

| SSConv       | Threshold | IDM in SA    | Params | FLOPs | Urban100     | Manga109     |
|--------------|-----------|--------------|--------|-------|--------------|--------------|
| $\times$     | -         | $\times$     | 141K   | 36G   | 32.22/0.9285 | 38.98/0.9770 |
| $\times$     | -         | $\checkmark$ | 141K   | 36G   | 32.31/0.9284 | 39.06/0.9793 |
| $\checkmark$ | 5%        | $\checkmark$ | 198K   | 42G   | 32.52/0.9312 | 39.32/0.9792 |
| $\checkmark$ | 10%       | $\checkmark$ | 198K   | 44G   | 32.50/0.9305 | 39.34/0.9792 |
| $\checkmark$ | 20%       | $\checkmark$ | 198K   | 47G   | 32.55/0.9313 | 39.36/0.9793 |
| $\checkmark$ | 50%       | $\checkmark$ | 198K   | 50G   | 32.58/0.9317 | 39.42/0.9799 |

Table 4. Ablation studies on different settings in IGRA.

| AR           | IDM          | Params | FLOPs | Urban100     | Manga109     |
|--------------|--------------|--------|-------|--------------|--------------|
| $\times$     | $\times$     | 167K   | 35G   | 32.30/0.9275 | 39.19/0.9774 |
| $\checkmark$ | $\times$     | 198K   | 42G   | 32.45/0.9297 | 39.26/0.9787 |
| $\checkmark$ | $\checkmark$ | 198K   | 42G   | 32.52/0.9312 | 39.32/0.9792 |

Table 5. Ablation studies on different settings in ARM.

information-oriented guidance.

**Effectiveness of the IGRA.** To assess the impact of the Information-Guided Resource Allocation module, we analyze its components in Table 4. First, incorporating the IDM into the self-attention mechanism provides a 0.08 dB performance gain with negligible computational overhead. As the core of IAFMNet, introducing SSConv brings a substantial improvement of 0.26 dB on Manga109 at the cost of only 6 GFLOPs, demonstrating its high efficiency. Finally, we study the sparse threshold. While higher thresholds yield slightly better absolute performance, the 5% threshold provides the best trade-off between performance and computational cost. This also supports our hypothesis that the most critical information is concentrated in a small fraction.

**Effectiveness of the ARM.** We validate our Affine Recalibration Module in Table 5. Although affine recalibration is a proven strategy in SISR [38], our study further shows the benefit of guiding it with our principled information prior. Simply adding the AR architecture provides a 0.07 dB gain on Manga109. Crucially, when the module is further guided by the IDM, it yields an additional 0.06 dB improvement on Manga109 with almost no increase in computational cost. This demonstrates the effectiveness of using the IDM as a guidance signal for fine-grained feature recalibration.

## 5. Conclusion

In this work, inspired by the spatially non-uniform information density in natural images, we introduce a novel information-aware feature modulation network (IAFMNet) for efficient SISR. We propose to estimate information density in an unsupervised manner by minimizing the Information Entropy Loss, thereby identifying informative regions corresponding to complex textures and edges. This is the first work to enhance the degraded features for SISR from the perspective of information density. We then leverage the estimated IDM to prioritize computational allocation and feature modulation in these informative areas. By adopting such an information-aware strategy, the proposed IAFMNet achieves a superior trade-off between reconstruction performance and computational cost compared with SOTA efficient SISR methods, demonstrating the feasibility and effectiveness of applying information theory to the SR task.

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